C PROGRAMMING LAB

Unit I • Unit II • Unit III

Nine lab programs with sample outputs

Sequence

Unit I: Basic input/output, area of a circle, average of three numbers
Unit II: Operator precedence, even or odd, positive/negative/zero
Unit III: Array values and addresses, sum of a one-dimensional array, matrix addition

UNIT I — BASIC INPUT/OUTPUT

1. Program to demonstrate basic input/output using different data types

Program:

```
#include <stdio.h>

int main()
{
    int age;
    float marks;
    double salary;
    char grade;

    printf("Enter your age: ");
    scanf("%d", &age);

    printf("Enter your marks: ");
    scanf("%f", &marks);

    printf("Enter your salary: ");
    scanf("%lf", &salary);

    printf("Enter your grade: ");
    scanf(" %c", &grade);

    printf("\\n--- Student Details ---\\n");
    printf("Age: %d\\n", age);
    printf("Marks: %.2f\\n", marks);
    printf("Salary: %.2f\\n", salary);
    printf("Grade: %c\\n", grade);

    return 0;
}
```

Sample Output:

```
Enter your age: 18
Enter your marks: 85.5
Enter your salary: 15000
Enter your grade: A

--- Student Details ---
Age: 18
Marks: 85.50
Salary: 15000.00
Grade: A
```

2. Program to calculate the area of a circle

Program:

```
#include <stdio.h>

int main()
{
    float radius, area;
    const float pi = 3.14f;

    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);

    area = pi * radius * radius;

    printf("Area of the circle = %.2f\\n", area);

    return 0;
}
```

Sample Output:

```
Enter the radius of the circle: 5
Area of the circle = 78.50
```

3. Program to calculate the average of three numbers

Program:

```
#include <stdio.h>

int main()
{
    float a, b, c, average;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);

    average = (a + b + c) / 3.0f;

    printf("Average = %.2f\\n", average);

    return 0;
}
```

Sample Output:

```
Enter three numbers: 10 20 30
Average = 20.00
```

UNIT II — DECISION MAKING AND LOOPING

1. Program to demonstrate the order of execution of operators in expressions

Program:

```
#include <stdio.h>

int main()
{
    int a = 10, b = 5, c = 2;
    int result;

    result = a + b * c;
    printf("a + b * c = %d\\n", result);

    result = (a + b) * c;
    printf("(a + b) * c = %d\\n", result);

    return 0;
}
```

Sample Output:

```
a + b * c = 20
(a + b) * c = 30
```

2. Program to check whether a given number is even or odd

Program:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0)
        printf("Number is Even\\n");
    else
```

```
        printf("Number is Odd\\n");

    return 0;
}
```

Sample Output:

```
Enter a number: 7
Number is Odd
```

3. Program to determine whether a number is positive, negative, or zero

Program:

```
#include <stdio.h>

int main()
{
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 0)
        printf("Number is Positive\\n");
    else if (num < 0)
        printf("Number is Negative\\n");
    else
        printf("Number is Zero\\n");

    return 0;
}
```

Sample Output:

```
Enter a number: -5
Number is Negative
```

UNIT III — ARRAYS AND FUNCTIONS

Note: Memory addresses vary between runs and computers, so the addresses below are shown as 0x... in the sample output.

1. Program to accept integer, float, and character arrays and display each value with its memory address

Program:

```
#include <stdio.h>

int main()
{
    int i;
    int numbers[3];
    float decimals[3];
    char letters[3];

    printf("Enter 3 integers: ");
    for (i = 0; i < 3; i++)
        scanf("%d", &numbers[i]);

    printf("Enter 3 float values: ");
    for (i = 0; i < 3; i++)
        scanf("%f", &decimals[i]);

    printf("Enter 3 characters: ");
    for (i = 0; i < 3; i++)
        scanf(" %c", &letters[i]);

    printf("\\nInteger array:\\n");
```

```
    for (i = 0; i < 3; i++)
        printf("Value = %d, Address = %p\\n",
               numbers[i], (void *)&numbers[i]);

    printf("\\nFloat array:\\n");
    for (i = 0; i < 3; i++)
        printf("Value = %.2f, Address = %p\\n",
               decimals[i], (void *)&decimals[i]);

    printf("\\nCharacter array:\\n");
    for (i = 0; i < 3; i++)
        printf("Value = %c, Address = %p\\n",
               letters[i], (void *)&letters[i]);

    return 0;
}
```

Sample Output:

```
Enter 3 integers: 10 20 30
Enter 3 float values: 1.5 2.5 3.5
Enter 3 characters: A B C

Integer array:
Value = 10, Address = 0x...
Value = 20, Address = 0x...
Value = 30, Address = 0x...

Float array:
Value = 1.50, Address = 0x...
Value = 2.50, Address = 0x...
Value = 3.50, Address = 0x...

Character array:
Value = A, Address = 0x...
Value = B, Address = 0x...
Value = C, Address = 0x...
```

2. Program to calculate the sum of all elements of a one-dimensional array

Program:

```
#include <stdio.h>

int main()
{
    int n, i, sum = 0;
    int arr[100];

    printf("Enter number of elements (1-100): ");
    scanf("%d", &n);

    if (n < 1 || n > 100)
    {
        printf("Invalid number of elements.\\n");
        return 1;
    }

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }

    printf("Sum of array elements = %d\\n", sum);

    return 0;
}
```

Sample Output:

```
Enter number of elements (1-100): 5
Enter 5 elements: 1 2 3 4 5
Sum of array elements = 15
```

3. Program to perform matrix addition using two-dimensional arrays

Program:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter rows and columns (max 10): ");
    scanf("%d %d", &rows, &cols);

    if (rows < 1 || rows > 10 || cols < 1 || cols > 10)
    {
        printf("Invalid matrix size.\\n");
        return 1;
    }

    printf("Enter elements of first matrix:\\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    printf("Enter elements of second matrix:\\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            scanf("%d", &b[i][j]);

    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            sum[i][j] = a[i][j] + b[i][j];

    printf("Resultant matrix:\\n");
    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
            printf("%d ", sum[i][j]);
        printf("\\n");
    }

    return 0;
}
```

Sample Output:

```
Enter rows and columns (max 10): 2 2
Enter elements of first matrix:
1 2
3 4
Enter elements of second matrix:
5 6
7 8
Resultant matrix:
6 8
10 12
```